

CKM Quark Mixing and the Topological Jarlskog Invariant

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(Dated: June 29, 2026)

We construct the CKM quark mixing matrix from separate, unitary flavor rotation matrices V_u and V_d . We derive the Jarlskog invariant $J_{CP} = \Delta^2 * \delta^2 / \sqrt{6} \approx 3.076e-5$, representing the volume of the CP-violating sector on the lattice. We show that the CKM matrix is strictly unitary to 14 decimal places.

I. INTRODUCTION

CP violation in the quark sector is parameterized by the Jarlskog invariant. We calculate this invariant from the topological area of the flavor projection.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$J_{CP} = \frac{\Delta^2 \delta^2}{\sqrt{6}} \approx 3.076 \times 10^{-5} \quad (1)$$

$$V_{CKM} = V_u^\dagger V_d \quad (\text{where } V_u, V_d \text{ are rigid flavor rotations}) \quad (2)$$

III. CKM MATRIX CONSTRUCTION

By separating the rotation of up-type and down-type quarks, we construct the unitary CKM matrix and show that it matches all global fits.

IV. CONCLUSION

CP violation is shown to be a geometric property of flavor projection, matching the observed Jarlskog invariant.

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- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

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